

Project Requirements – Part 1: Group Management System

1. Overview

We are building a **Group Management System (GMS)**.

This is a simple system to manage:

- **Groups** (e.g., WhatsApp, Telegram, Facebook, etc.)
- **Members** (people in those groups)
- **Messages** (conversations inside groups)

The system should allow us to:

- Create, update, and delete groups.
- Add members to groups.
- Send and store messages from members inside groups.
- Track which member belongs to which group.

2. Core Features (Part 1 Scope)

1. Authentication

- Simple login system for administrators (username + password).
- Only logged-in users can manage data.

2. Group Management

- Create, update, delete groups.
- Each group belongs to a platform (WhatsApp, Telegram, etc.).
- Groups can be **Group / Channel / Broadcast** type.
- Groups have statuses:
 - 0 → inactive
 - 1 → active (default)
 - 5 → deleted

3. Member Management

- Add, update, delete members.
- Each member can have:
 - Full name
 - Phone number
 - Email
 - Username

- Multiple platforms they belong to

4. Message Management

- Add, update, delete messages.
- Messages belong to a **group**.
- Each message has a **sender** (member).
- Messages can have text and/or media (image, video, etc.).

5. Group Memberships

- Define which members belong to which groups.
- Mark members as **admins** inside a group.

3. Database Tables

Groups

- `id` (PK)
- `name`
- `platform` (WhatsApp, Telegram, etc.)
- `group_type` (Group / Channel / Broadcast)
- `added_time`
- `last_updated_time`
- `status` (enum → 0 = inactive, 1 = active, 5 = deleted)

Members

- `id` (PK)
- `full_name`
- `phone_number`
- `email`
- `username`
- `platforms` (can store multiple platforms)
- `added_time`
- `last_updated_time`
- `status` (enum → 0 = inactive, 1 = active, 5 = deleted)

Messages

- `id` (PK)
- `text_body`
- `has_media` (boolean)

- `media_url`
- `group_id` (FK → Groups)
- `sender_id` (FK → Members)
- `added_time`
- `last_updated_time`
- `status` (enum → 0 = inactive, 1 = active, 5 = deleted)

Group Memberships

- `id` (PK)
- `group_id` (FK → Groups)
- `member_id` (FK → Members)
- `is_admin` (boolean)
- `added_time`
- `last_updated_time`
- `status` (enum → 0 = inactive, 1 = active, 5 = deleted)